

INTAKE-ARBITER -- FREE-COOLING DECISION REPORT

Data centres over-cool, continuously, because nobody can promise them the hours ahead. A chiller plant needs hours of notice to change mode, and a thermometer only ever reports NOW -- so the mechanical chillers keep running through hours that outside air could have cooled for free. FortyGuard closes that gap with heat intelligence 2 m above the ground, the height a ground-mounted condenser actually breathes. This agent turns that forecast into an hour-by-hour schedule carrying a calibrated safety bound.

THE SITE

Location Google, IA -- station KCBF, America/Chicago
Committed pair OSM 1073720220 -> 1073720216
Google / Google
Facade gap 188.3 m between the two halls
Weather record 42,455 real hourly records from KCBF
Plume physics 576 steady-state solves on this site's own footprints; worst intake rise 0.4285 C at 175 deg

FortyGuard data: weather and geometry are this site's own; the four measured FortyGuard level offsets and the N-26 coverage record are Ashburn's, because only Ashburn has forecast/outcome day pairs. Quote the hours for this site; quote the coverage for Ashburn.

THIS REPORT IS A SNAPSHOT, NOT THE LIVE PAGE

The interface recomputes the decision for whatever configuration a reader selects. This file was generated at build time for the ONE configuration named below, chosen because it is the configuration in which this site's agent does the most while still changing mode at least once. If the numbers on screen differ from the numbers here, the configuration differs -- compare the two lists before concluding anything else.

Day 2022-04-12 (crossing)
Plant limit 21.0 C
Notice required 3 h
Forecast skill 0.50 relative to persistence
Level anchor sensor
Condenser bank longest facade
Switch budget 2 mode changes per day
Minimum dwell 3 h
Max dew point 15.0 C (Green Grid WP#46 p.6)

WHAT THE AGENT DID ON THIS DAY

The agent ran free cooling for 3 of 24 hours with 2 mode changes. The reactive incumbent took 15 hours -- 12 more -- but declared 1 unsafe hour against the agent's 0.5 hours were safe but still ran chillers, because the schedule could not afford them.

Agent 3 of 24 hours free cooling, 2 mode change(s), 0 unsafe hour(s)
Incumbent 15 of 24 hours free cooling, 2 mode change(s), 1 unsafe hour(s)
5 hour(s) passed every gate and still ran chillers, because the schedule could not afford them

EVERY HOUR, AND THE REASON FOR IT

hh	mode	safe	binding	bound	limit	actual	margin
00	FREE	yes	-	8.076	21.0	8.002	0.922
01	FREE	yes	-	8.116	21.0	8.001	0.880
02	FREE	yes	-	8.125	21.0	8.001	0.833
03	mechanical	no	refusal	9.341	21.0	10.001	0.845
04	mechanical	yes	-	9.101	21.0	10.001	0.697
05	mechanical	yes	-	9.574	21.0	10.000	1.116
06	mechanical	no	refusal	10.643	21.0	10.000	1.057

07	mechanical	no	refusal	11.634	21.0	11.000	1.154
08	mechanical	yes	-	13.089	21.0	12.000	1.305
09	mechanical	no	refusal	14.146	21.0	11.004	1.988
10	mechanical	no	refusal	15.699	21.0	12.003	1.961
11	mechanical	no	refusal	16.766	21.0	13.025	1.940
12	mechanical	no	refusal	16.955	21.0	15.025	1.711
13	mechanical	no	refusal	19.131	21.0	20.005	1.559
14	mechanical	no	refusal	23.160	21.0	28.029	1.497
15	mechanical	no	refusal	24.627	21.0	31.003	0.922
16	mechanical	no	refusal	27.136	21.0	32.003	0.971
17	mechanical	no	refusal	30.248	21.0	31.003	1.061
18	mechanical	no	refusal	30.631	21.0	30.005	1.075
19	mechanical	no	refusal	29.748	21.0	27.258	1.495
20	mechanical	no	dry-bulb	22.575	21.0	14.000	1.609
21	mechanical	no	dry-bulb	22.031	21.0	14.000	1.504
22	mechanical	yes	switch budget	20.081	21.0	13.000	1.309
23	mechanical	yes	switch budget	12.563	21.0	11.000	1.043

bound = the agent's 90 %-nominal upper bound on intake air, forecast plus margin
actual = what the intake temperature turned out to be, from the station record
margin = group-conditional forecast error for that hour of day, plus plume spread

THE REASONING, HOUR BY HOUR

00:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 8.076 C, which is 12.924 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 0.922 C, and the margin is measured, not chosen: 0.846 C of group-conditional forecast error for this hour of day, plus 0.0764 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

01:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 8.116 C, which is 12.884 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 0.880 C, and the margin is measured, not chosen: 0.764 C of group-conditional forecast error for this hour of day, plus 0.1163 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

02:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 8.125 C, which is 12.875 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 0.833 C, and the margin is measured, not chosen: 0.708 C of group-conditional forecast error for this hour of day, plus 0.1249 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

03:00 MECHANICAL [refusal]

Mechanical, and the reason is a REFUSAL rather than a temperature. At this wind bearing a building sits between the condensers and the air intake. The dispersion model has no representation of a building standing in the flow, so any number it produced here would be meaningless. The agent declines to certify the hour instead of returning a figure it cannot stand behind.

04:00 MECHANICAL

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 9.100 C against a 21.0 C limit. The schedule is what forbids it: no relaxation of either the switch budget or the dwell would free it, so the surrounding hours are worth more. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

05:00 MECHANICAL

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 9.574 C against a 21.0 C limit. The schedule is what forbids it: no relaxation of either the

switch budget or the dwell would free it, so the surrounding hours are worth more. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

06:00 MECHANICAL [refusal]

Mechanical, and the reason is a REFUSAL rather than a temperature. At this wind bearing a building sits between the condensers and the air intake. The dispersion model has no representation of a building standing in the flow, so any number it produced here would be meaningless. The agent declines to certify the hour instead of returning a figure it cannot stand behind.

07:00 MECHANICAL [refusal]

Mechanical, and the reason is a REFUSAL rather than a temperature. At this wind bearing a building sits between the condensers and the air intake. The dispersion model has no representation of a building standing in the flow, so any number it produced here would be meaningless. The agent declines to certify the hour instead of returning a figure it cannot stand behind.

08:00 MECHANICAL

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 13.089 C against a 21.0 C limit. The schedule is what forbids it: no relaxation of either the switch budget or the dwell would free it, so the surrounding hours are worth more. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

09:00 MECHANICAL [refusal]

Mechanical, and the reason is a REFUSAL rather than a temperature. At this wind bearing a building sits between the condensers and the air intake. The dispersion model has no representation of a building standing in the flow, so any number it produced here would be meaningless. The agent declines to certify the hour instead of returning a figure it cannot stand behind.

10:00 MECHANICAL [refusal]

Mechanical, and the reason is a REFUSAL rather than a temperature. At this wind bearing a building sits between the condensers and the air intake. The dispersion model has no representation of a building standing in the flow, so any number it produced here would be meaningless. The agent declines to certify the hour instead of returning a figure it cannot stand behind.

11:00 MECHANICAL [refusal]

Mechanical, and the reason is a REFUSAL rather than a temperature. At this wind bearing a building sits between the condensers and the air intake. The dispersion model has no representation of a building standing in the flow, so any number it produced here would be meaningless. The agent declines to certify the hour instead of returning a figure it cannot stand behind.

12:00 MECHANICAL [refusal]

Mechanical, and the reason is a REFUSAL rather than a temperature. At this wind bearing a building sits between the condensers and the air intake. The dispersion model has no representation of a building standing in the flow, so any number it produced here would be meaningless. The agent declines to certify the hour instead of returning a figure it cannot stand behind.

13:00 MECHANICAL [refusal]

Mechanical, and the reason is a REFUSAL rather than a temperature. At this wind bearing a building sits between the condensers and the air intake. The dispersion model has no representation of a building standing in the flow, so any number it produced here would be meaningless. The agent declines to certify the hour instead of returning a figure it cannot stand behind.

14:00 MECHANICAL [refusal]

Mechanical, and the reason is a REFUSAL rather than a temperature. At this wind bearing a building sits between the condensers and the air intake. The dispersion model has no representation of a building standing in the flow, so any number it produced here

would be meaningless. The agent declines to certify the hour instead of returning a figure it cannot stand behind.

15:00 MECHANICAL [refusal]

Mechanical, and the reason is a REFUSAL rather than a temperature. At this wind bearing a building sits between the condensers and the air intake. The dispersion model has no representation of a building standing in the flow, so any number it produced here would be meaningless. The agent declines to certify the hour instead of returning a figure it cannot stand behind.

16:00 MECHANICAL [refusal]

Mechanical, and the reason is a REFUSAL rather than a temperature. At this wind bearing a building sits between the condensers and the air intake. The dispersion model has no representation of a building standing in the flow, so any number it produced here would be meaningless. The agent declines to certify the hour instead of returning a figure it cannot stand behind.

17:00 MECHANICAL [refusal]

Mechanical, and the reason is a REFUSAL rather than a temperature. At this wind bearing a building sits between the condensers and the air intake. The dispersion model has no representation of a building standing in the flow, so any number it produced here would be meaningless. The agent declines to certify the hour instead of returning a figure it cannot stand behind.

18:00 MECHANICAL [refusal]

Mechanical, and the reason is a REFUSAL rather than a temperature. At this wind bearing a building sits between the condensers and the air intake. The dispersion model has no representation of a building standing in the flow, so any number it produced here would be meaningless. The agent declines to certify the hour instead of returning a figure it cannot stand behind.

19:00 MECHANICAL [refusal]

Mechanical, and the reason is a REFUSAL rather than a temperature. At this wind bearing a building sits between the condensers and the air intake. The dispersion model has no representation of a building standing in the flow, so any number it produced here would be meaningless. The agent declines to certify the hour instead of returning a figure it cannot stand behind.

20:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 22.575 C against a 21.0 C limit, so it fails by 1.575 C. It would take a limit 1.575 C higher, or a bound 1.575 C tighter, to change this hour.

21:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 22.031 C against a 21.0 C limit, so it fails by 1.031 C. It would take a limit 1.031 C higher, or a bound 1.031 C tighter, to change this hour.

22:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 20.081 C against a 21.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

23:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 12.563 C against a 21.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

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